

Multiclass Credit Risk Classification DATA Report

1. Task overview

This project addresses a **multiclass classification** problem in the context of consumer lending. The goal is to predict the **credit risk grade** assigned to a loan applicant using information available at (or before) loan issuance.

Prediction target

- **Target label:** `grade` $\in \{A, B, C, D, E, F, G\}$
where **A** corresponds to lower risk and **G** to higher risk (7 classes).
- **Numeric encoding:** `grade_idx` $\in \{0, 1, 2, 3, 4, 5, 6\}$ mapping A -> 0, ..., G -> 6.

This satisfies the assignment's multiclass requirement (≥ 3 classes).

2. Data origin

The dataset originates from **LendingClub loan statistics** (historical loan application and credit attributes). The raw file used for preprocessing was:

- `LendingClub_2007_to_2018Q4.csv`
- https://bigblue.depaul.edu/jlee141/econdata/LendingClub_LoanData/

The raw dataset contains millions of loans and a large number of columns, including both pre-issuance application attributes and post-issuance performance fields (payments, recoveries, delinquencies after issuance, hardship programs, etc.).

3. Final feature set

The final modeling dataset contains **24 columns**: 2 labels + 22 features.

3.1 Labels

- `grade` (A–G)
- `grade_idx` (0–6)

3.2 Features

Categorical features (one-hot encoding required for most models)

1. `purpose` — loan purpose category
2. `home_ownership` — borrower's housing status (RENT/OWN/MORTGAGE/...)
3. `verification_status` — income verification status
4. `addr_state` — borrower's state

Numeric features (standardized for PCA/LogReg/SVM/MLP; used directly for NB with Gaussian assumption)

1. `loan_amnt` — requested loan amount
2. `term` — loan term in months (36 or 60)
3. `emp_length` — employment length (0–10)
4. `annual_inc` — annual income

5. `dti` — debt-to-income ratio
 6. `delinq_2yrs` — delinquencies in last 2 years
 7. `inq_last_6mths` — inquiries in last 6 months
 8. `open_acc` — number of open credit lines
 9. `total_acc` — total credit lines
 10. `pub_rec` — derogatory public records
 11. `revol_bal` — revolving credit balance
 12. `revol_util` — revolving utilization rate (%)
 13. `collections_12_mths_ex_med` — collections excluding medical (12 months)
 14. `acc_now_delinq` — accounts currently delinquent
 15. `pub_rec_bankruptcies` — bankruptcies count
 16. `tax_liens` — tax liens count
 17. `credit_history_months` — derived: months between earliest credit line and issue date
 18. `fico_avg` — derived: average of FICO low/high ranges
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4. Data splitting strategy

The dataset was split into:

- **Train:** 70%
 - **Dev:** 15%
 - **Test:** 15%
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5. Dataset statistics

5.1 Training set

- Shape: **59,619 rows × 24 columns**
- Per-class counts: **8,517 for each grade A–G**
- Class proportions: **0.1429 per class**

5.2 Development set

- Shape: **12,775 rows × 24 columns**
- Per-class counts: **1,825 for each grade A–G**
- Class proportions: **0.1429 per class**

5.3 Test set

- Shape: **12,782 rows × 24 columns**
- Per-class counts: **1,826 for each grade A–G**
- Class proportions: **0.1429 per class**